

Ex No: 1a)

Date:

INSTALLATION AND CONFIGURATION OF LINUX

Aim:

To install and configure Linux operating system in a Virtual Machine.

Installation/Configuration Steps:

1. Install the required packages for virtualization
dnf install xen virt-manager qemu libvirt
2. Configure xend to start up on boot
systemctl enable virt-manager.service
3. Reboot the machine
Reboot
4. Create Virtual machine by first running virt-manager
virt-manager &
5. Click on File and then click to connect to localhost
6. In the base menu, right click on the localhost(QEMU) to create a new VM 7. Select

user1@localhost:~



```
[user1@localhost ~]$ mkdir rec
[user1@localhost ~]$ cd rec
[user1@localhost rec]$ cd ..
[user1@localhost ~]$ rmdir rec
[user1@localhost ~]$ ls
Desktop Documents Downloads fold2 fold3 Music Pictures Public sample Templates Videos
[user1@localhost ~]$ touch os.txt
[user1@localhost ~]$ vi os.txt
[user1@localhost ~]$ touch os2.txt
[user1@localhost ~]$ cp os.txt os2.txt
[user1@localhost ~]$ cat os2.txt
This is Linux operating system.
[user1@localhost ~]$ rm os2.txt
[user1@localhost ~]$ ls
Desktop Documents Downloads fold2 fold3 Music os.txt Pictures Public sample Templates Videos
[user1@localhost ~]$ cd ..
[user1@localhost home]$ ls
user1
[user1@localhost home]$ cd user1
[user1@localhost ~]$ ls
Desktop Documents Downloads fold2 fold3 Music os.txt Pictures Public sample Templates Videos
[user1@localhost ~]$ mkdir rec
[user1@localhost ~]$ mv os.txt rec
[user1@localhost ~]$ cd rec
[user1@localhost rec]$ ls
os.txt
[user1@localhost rec]$ file os.txt
os.txt: ASCII text
[user1@localhost rec]$ wc os.txt
 1  5 33 os.txt
[user1@localhost rec]$ ls>~
bash: /home/user1: Is a directory
[user1@localhost rec]$ ls>user1
[user1@localhost rec]$ cd ..
[user1@localhost ~]$ ls
Desktop Documents Downloads fold2 fold3 Music Pictures Public rec sample Templates Videos
[user1@localhost ~]$ who | date
Monday 19 January 2026 05:53:35 AM IST
[user1@localhost ~]$ who | tee rec
tee: rec: Is a directory
user1  seat0      2026-01-19 05:27 (login screen)
user1  tty2       2026-01-19 05:27 (tty2)
```

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user1@localhost:~/rec

```
user1  tty2      2026-01-19 05:27 (tty2)
[user1@localhost ~]$ ls r**
os.txt  user1
[user1@localhost ~]$ gg
bash: gg: command not found...
[user1@localhost ~]$ who ; date
user1    seat0      2026-01-19 05:27 (login screen)
user1    tty2      2026-01-19 05:27 (tty2)
Monday 19 January 2026 05:56:29 AM IST
[user1@localhost ~]$ mkdir test && cd test
[user1@localhost test]$ cd .. || echo "Directory changed"
[user1@localhost ~]$ ls
Desktop  Documents  Downloads  fold2  fold3  Music  Pictures  Public  rec  sample  Templates  test  Videos
[user1@localhost ~]$ cd rec
[user1@localhost rec]$ ls
os.txt  user1
[user1@localhost rec]$ os.txt
bash: os.txt: command not found...
[user1@localhost rec]$ head os.txt
This is Linux operating system.
[user1@localhost rec]$ tail  os.txt
This is Linux operating system.
[user1@localhost rec]$ tail -4 os.txt
This is Linux operating system.
[user1@localhost rec]$ ls | more
os.txt
user1
[user1@localhost rec]$ grep "is" os.txt
This is Linux operating system.
[user1@localhost rec]$ sort os.txt
This is Linux operating system.
[user1@localhost rec]$ sort -r os.txt
This is Linux operating system.
[user1@localhost rec]$ nl os.txt
     1  This is Linux operating system.
[user1@localhost rec]$ cut -c 1-8 os.txt
This is
[user1@localhost rec]$ free -h
               total        used        free      shared  buff/cache   available
Mem:            1.7Gi         902Mi        280Mi        16Mi        695Mi        804Mi
Swap:           2.0Gi          3.0Mi        2.0Gi
```

user1@localhost:~/rec



```
/dev/sda1      960M  439M  522M  46% /boot
tmpfs          171M  100K  171M   1% /run/user/1000
[user1@localhost rec]$ ping google.com
PING google.com (172.217.24.142) 56(84) bytes of data.
64 bytes from syd09s06-in-f14.1e100.net (172.217.24.142): icmp_seq=1 ttl=118 time=7.81 ms
64 bytes from syd09s06-in-f14.1e100.net (172.217.24.142): icmp_seq=2 ttl=118 time=8.61 ms
64 bytes from nrt20s01-in-f142.1e100.net (172.217.24.142): icmp_seq=3 ttl=118 time=9.14 ms
64 bytes from nrt20s01-in-f14.1e100.net (172.217.24.142): icmp_seq=4 ttl=118 time=8.29 ms
64 bytes from lcmaaa-av-in-f14.1e100.net (172.217.24.142): icmp_seq=5 ttl=118 time=7.31 ms
64 bytes from nrt20s01-in-f14.1e100.net (172.217.24.142): icmp_seq=6 ttl=118 time=7.02 ms
64 bytes from lcmaaa-av-in-f14.1e100.net (172.217.24.142): icmp_seq=7 ttl=118 time=7.14 ms
64 bytes from nrt20s01-in-f142.1e100.net (172.217.24.142): icmp_seq=8 ttl=118 time=9.63 ms
64 bytes from lcmaaa-av-in-f14.1e100.net (172.217.24.142): icmp_seq=9 ttl=118 time=7.63 ms
64 bytes from syd09s06-in-f14.1e100.net (172.217.24.142): icmp_seq=10 ttl=118 time=10.9 ms
64 bytes from syd09s06-in-f14.1e100.net (172.217.24.142): icmp_seq=11 ttl=118 time=13.1 ms
64 bytes from nrt20s01-in-f142.1e100.net (172.217.24.142): icmp_seq=12 ttl=118 time=8.86 ms
64 bytes from lcmaaa-av-in-f14.1e100.net (172.217.24.142): icmp_seq=13 ttl=118 time=8.37 ms
64 bytes from nrt20s01-in-f14.1e100.net (172.217.24.142): icmp_seq=14 ttl=118 time=8.74 ms
64 bytes from nrt20s01-in-f14.1e100.net (172.217.24.142): icmp_seq=15 ttl=118 time=9.42 ms
64 bytes from nrt20s01-in-f14.1e100.net (172.217.24.142): icmp_seq=16 ttl=118 time=8.48 ms
64 bytes from syd09s06-in-f14.1e100.net (172.217.24.142): icmp_seq=17 ttl=118 time=7.70 ms
64 bytes from nrt20s01-in-f142.1e100.net (172.217.24.142): icmp_seq=18 ttl=118 time=10.6 ms
64 bytes from nrt20s01-in-f142.1e100.net (172.217.24.142): icmp_seq=19 ttl=118 time=8.63 ms
64 bytes from lcmaaa-av-in-f14.1e100.net (172.217.24.142): icmp_seq=20 ttl=118 time=8.58 ms
64 bytes from nrt20s01-in-f142.1e100.net (172.217.24.142): icmp_seq=21 ttl=118 time=10.4 ms
64 bytes from syd09s06-in-f14.1e100.net (172.217.24.142): icmp_seq=22 ttl=118 time=13.0 ms
64 bytes from nrt20s01-in-f14.1e100.net (172.217.24.142): icmp_seq=23 ttl=118 time=9.08 ms

64 bytes from lcmaaa-av-in-f14.1e100.net (172.217.24.142): icmp_seq=24 ttl=118 time=8.26 ms
64 bytes from nrt20s01-in-f142.1e100.net (172.217.24.142): icmp_seq=25 ttl=118 time=7.24 ms
64 bytes from nrt20s01-in-f14.1e100.net (172.217.24.142): icmp_seq=26 ttl=118 time=7.64 ms
64 bytes from lcmaaa-av-in-f14.1e100.net (172.217.24.142): icmp_seq=27 ttl=118 time=8.17 ms
64 bytes from syd09s06-in-f14.1e100.net (172.217.24.142): icmp_seq=28 ttl=118 time=7.15 ms
64 bytes from lcmaaa-av-in-f14.1e100.net (172.217.24.142): icmp_seq=29 ttl=118 time=9.23 ms
64 bytes from nrt20s01-in-f142.1e100.net (172.217.24.142): icmp_seq=30 ttl=118 time=11.1 ms
64 bytes from syd09s06-in-f14.1e100.net (172.217.24.142): icmp_seq=31 ttl=118 time=8.31 ms
64 bytes from lcmaaa-av-in-f14.1e100.net (172.217.24.142): icmp_seq=32 ttl=118 time=9.52 ms
^C
--- google.com ping statistics ---
32 packets transmitted, 32 received, 0% packet loss, time 31067ms
rtt min/avg/max/mdev = 7.015/8.908/13.142/1.514 ms
[user1@localhost rec]$
```

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Jan 19 06:02

user1@localhost:~/rec

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```
init: is
[user1@localhost rec]$ free -h
              total        used         free       shared  buff/cache   available
Mem:          1.7Gi         902Mi        280Mi         16Mi         695Mi         804Mi
Swap:          2.0Gi         3.0Mi         2.0Gi

[user1@localhost rec]$ ps
  PID TTY          TIME CMD
 2510 pts/0    00:00:00 bash
 3013 pts/0    00:00:00 ps

[user1@localhost rec]$ vmstat
procs -----memory----- --swap--  -----io----- -system--  -----cpu-----
 r  b   swpd   free   buff  cache   si   so    bi    bo   in   cs  us  sy  id  wa  st
 1   0    3352 287712   3264 708716    0    2   377   108  113 349  0  3 97  0  0

[user1@localhost rec]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        821M   0   821M   0% /dev
tmpfs            854M   0   854M   0% /dev/shm
tmpfs            342M  5.5M  337M   2% /run
/dev/mapper/cs-root 19G  5.2G   14G  29% /
/dev/sdal        960M  439M  522M  46% /boot
tmpfs            171M 100K  171M   1% /run/user/1000

[user1@localhost rec]$ ping google.com
PING google.com (172.217.24.142) 56(84) bytes of data.
54 bytes from syd09s06-in-f14.1e100.net (172.217.24.142): icmp_seq=1 ttl=118 time=7.81 ms
54 bytes from syd09s06-in-f14.1e100.net (172.217.24.142): icmp_seq=2 ttl=118 time=8.61 ms
54 bytes from nrt20s01-in-f142.1e100.net (172.217.24.142): icmp_seq=3 ttl=118 time=9.14 ms
54 bytes from nrt20s01-in-f14.1e100.net (172.217.24.142): icmp_seq=4 ttl=118 time=8.29 ms
54 bytes from lcmaaa-av-in-f14.1e100.net (172.217.24.142): icmp_seq=5 ttl=118 time=7.31 ms
54 bytes from nrt20s01-in-f14.1e100.net (172.217.24.142): icmp_seq=6 ttl=118 time=7.02 ms
54 bytes from lcmaaa-av-in-f14.1e100.net (172.217.24.142): icmp_seq=7 ttl=118 time=7.14 ms
54 bytes from nrt20s01-in-f142.1e100.net (172.217.24.142): icmp_seq=8 ttl=118 time=9.63 ms
54 bytes from lcmaaa-av-in-f14.1e100.net (172.217.24.142): icmp_seq=9 ttl=118 time=7.63 ms
54 bytes from syd09s06-in-f14.1e100.net (172.217.24.142): icmp_seq=10 ttl=118 time=10.9 ms
54 bytes from syd09s06-in-f14.1e100.net (172.217.24.142): icmp_seq=11 ttl=118 time=13.1 ms
54 bytes from nrt20s01-in-f142.1e100.net (172.217.24.142): icmp_seq=12 ttl=118 time=8.86 ms
54 bytes from lcmaaa-av-in-f14.1e100.net (172.217.24.142): icmp_seq=13 ttl=118 time=8.37 ms
54 bytes from nrt20s01-in-f14.1e100.net (172.217.24.142): icmp_seq=14 ttl=118 time=8.74 ms
54 bytes from nrt20s01-in-f14.1e100.net (172.217.24.142): icmp_seq=15 ttl=118 time=9.42 ms
54 bytes from nrt20s01-in-f14.1e100.net (172.217.24.142): icmp_seq=16 ttl=118 time=8.48 ms
54 bytes from syd09s06-in-f14.1e100.net (172.217.24.142): icmp_seq=17 ttl=118 time=7.70 ms
54 bytes from nrt20s01-in-f142.1e100.net (172.217.24.142): icmp_seq=18 ttl=118 time=10.6 ms
54 bytes from nrt20s01-in-f142.1e100.net (172.217.24.142): icmp_seq=19 ttl=118 time=8.63 ms
54 bytes from lcmaaa-av-in-f14.1e100.net (172.217.24.142): icmp_seq=20 ttl=118 time=8.58 ms
```


user1@localhost:~/rec



```
64 bytes from nrt20s01-in-f142.1e100.net (172.217.24.142): icmp_seq=30 ttl=118 time=11.1 ms
64 bytes from syd09s06-in-f14.1e100.net (172.217.24.142): icmp_seq=31 ttl=118 time=8.31 ms
64 bytes from lcmaaa-av-in-f14.1e100.net (172.217.24.142): icmp_seq=32 ttl=118 time=9.52 ms
^C
```

```
--- google.com ping statistics ---
```

```
32 packets transmitted, 32 received, 0% packet loss, time 31067ms
rtt min/avg/max/mdev = 7.015/8.908/13.142/1.514 ms
```

```
[user1@localhost rec]$ ifconfig
```

```
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::a00:27ff:fe3a:b8d5 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:3a:b8:d5 txqueuelen 1000 (Ethernet)
    RX packets 239 bytes 26312 (25.6 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 286 bytes 25993 (25.3 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
```

```
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 18 bytes 2112 (2.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 18 bytes 2112 (2.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
[user1@localhost rec]$ date
```

```
Monday 19 January 2026 06:03:36 AM IST
```

```
[user1@localhost rec]$ echo "This is Linux OS"
```

```
This is Linux OS
```

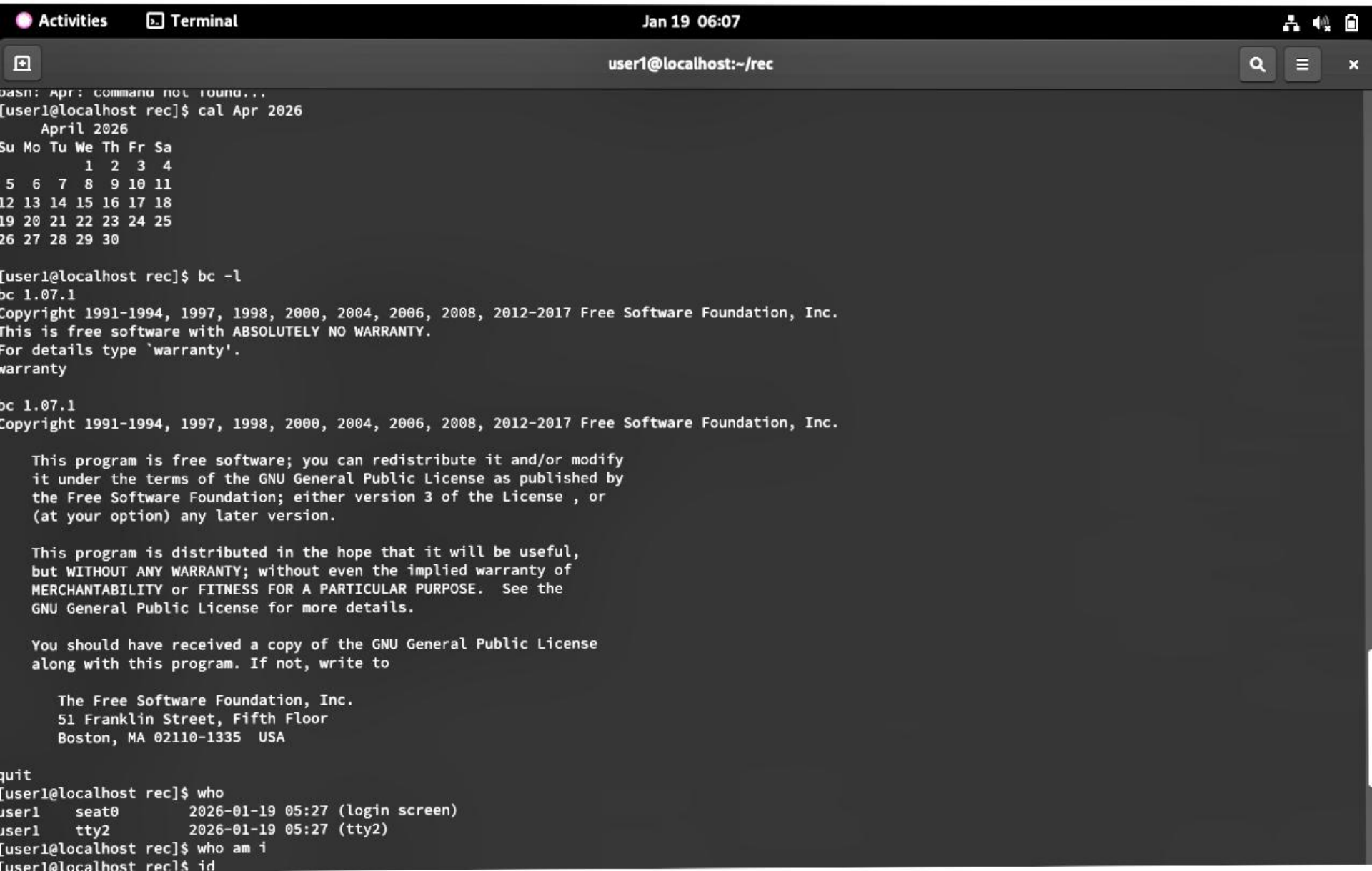
```
[user1@localhost rec]$ Apr 2026
```

```
bash: Apr: command not found...
```

```
[user1@localhost rec]$ cal Apr 2026
```

```
    April 2026
Su Mo Tu We Th Fr Sa
                1  2  3  4
 5  6  7  8  9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30
```

```
[user1@localhost rec]$
```



bash: Apr: command not found...

[user1@localhost rec]\$ cal Apr 2026

```
April 2026
Su Mo Tu We Th Fr Sa
                1  2  3  4
 5  6  7  8  9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30
```

[user1@localhost rec]\$ bc -l

```
bc 1.07.1
Copyright 1991-1994, 1997, 1998, 2000, 2004, 2006, 2008, 2012-2017 Free Software Foundation, Inc.
This is free software with ABSOLUTELY NO WARRANTY.
For details type `warranty'.
warranty
```

```
bc 1.07.1
Copyright 1991-1994, 1997, 1998, 2000, 2004, 2006, 2008, 2012-2017 Free Software Foundation, Inc.
```

```
This program is free software; you can redistribute it and/or modify
it under the terms of the GNU General Public License as published by
the Free Software Foundation; either version 3 of the License , or
(at your option) any later version.
```

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This program is distributed in the hope that it will be useful,
but WITHOUT ANY WARRANTY; without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  See the
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along with this program. If not, write to
```

```
The Free Software Foundation, Inc.
51 Franklin Street, Fifth Floor
Boston, MA 02110-1335  USA
```

quit

[user1@localhost rec]\$ who

```
user1    seat0        2026-01-19 05:27 (login screen)
user1    tty2         2026-01-19 05:27 (tty2)
```

[user1@localhost rec]\$ who am i

[user1@localhost rec]\$ id

user1@localhost:~/rec

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along with this program. If not, write to

The Free Software Foundation, Inc.
51 Franklin Street, Fifth Floor
Boston, MA 02110-1335 USA

quit

```
[user1@localhost rec]$ who
user1  seat0      2026-01-19 05:27 (login screen)
user1  tty2        2026-01-19 05:27 (tty2)
```

```
[user1@localhost rec]$ who am i
```

```
[user1@localhost rec]$ id
```

```
uid=1000(user1) gid=1000(user1) groups=1000(user1) context=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
```

```
[user1@localhost rec]$ tty
```

```
/dev/pts/0
```

```
[user1@localhost rec]$ touch os.txt
```

```
[user1@localhost rec]$ ps
```

PID	TTY	TIME	CMD
2510	pts/0	00:00:00	bash
3119	pts/0	00:00:00	ps

```
[user1@localhost rec]$ unname Linux
```

```
bash: unname: command not found...
```

```
Similar command is: 'uname'
```

```
[user1@localhost rec]$ uname Linux
```

```
uname: extra operand 'Linux'
```

```
Try 'uname --help' for more information.
```

```
[user1@localhost rec]$ uname -m
```

```
x86_64
```

```
[user1@localhost rec]$ uname -n
```

```
localhost.localdomain
```

```
[user1@localhost rec]$ uname -a
```

```
Linux localhost.localdomain 5.14.0-654.el9.x86_64 #1 SMP PREEMPT_DYNAMIC Fri Dec 19 08:34:59 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
```

```
[user1@localhost rec]$ S
```